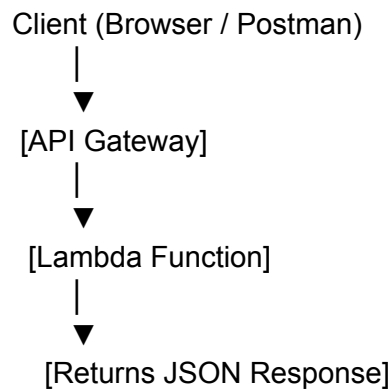

Lab Title: Create a RESTful API using AWS API Gateway & Lambda

Lab Objective

Build a REST API that triggers an AWS Lambda function which returns a JSON response.

Architecture Overview



Prerequisites

- AWS account
 - IAM user with permissions for API Gateway, Lambda, CloudWatch
 - Basic knowledge of AWS Console
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Step-by-Step Instructions

Step 1: Create a Lambda Function

1. Go to AWS Console → **Lambda**
2. Click **Create function**
3. Choose:
 - **Author from scratch**
 - Function name: `HelloWorldFunction`
 - Runtime: **Python 3.12** or **Node.js 18.x**
 - Permissions: **Create a new role with basic Lambda permissions**
4. Click **Create Function**
5. Replace the default code with:

For Python:

```
def lambda_handler(event, context):  
    return {  
        "statusCode": 200,  
        "body": "Hello from Lambda!",  
        "headers": {  
            "Content-Type": "application/json"  
        }  
    }  
}
```

For Node.js:

```
exports.handler = async (event) => {  
    return {  
        statusCode: 200,  
        body: JSON.stringify("Hello from Lambda!"),  
        headers: {  
            "Content-Type": "application/json"  
        }  
    };  
};
```

6. Click **Deploy**

✅ Step 2: Create REST API in API Gateway

1. Go to **Amazon API Gateway**
2. Choose **REST API** → Click **Build**
3. Choose:
 - API name: **HelloAPI**
 - Endpoint type: **Regional**
4. Click **Create API**

✅ Step 3: Create Resource and Method

1. Under **Resources**, click **Actions** → **Create Resource**
 - Resource Name: **hello**
 - Resource Path: **/hello**
 - Click **Create Resource**
2. Select **/hello**, click **Actions** → **Create Method** → Choose **GET**
3. Integration type: **Lambda Function**
4. Lambda Region: Choose correct region
5. Lambda Function: **HelloWorldFunction**
6. Click **Save**, then **OK** to give API Gateway permissions

✅ Step 4: Deploy API

1. Click **Actions** → **Deploy API**

2. Deployment stage:

- Choose **New Stage**
- Stage name: **dev**
- Click **Deploy**

Note the **Invoke URL**:

`https://<api-id>.execute-api.<region>.amazonaws.com/dev/hello`

3.

Step 5: Test API

Use **Postman**, **curl**, or browser:

`curl https://<api-id>.execute-api.<region>.amazonaws.com/dev/hello`

 Expected Output:

"Hello from Lambda!"

Optional: Enable Logging

1. In API Gateway → Settings → Enable **CloudWatch Logs**
 2. View logs in **CloudWatch** → **Log Groups** → `/aws/apigateway/`
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Extension Ideas (Advanced Labs)

- Add **Cognito authentication**
 - Use **API keys & usage plans**
 - Create **custom domain name**
 - Add **Stage variables**
 - Enable **Throttling & Rate limiting**
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